TETOUAN CITY POWER CONSUMPTION FORECASTING

Group Task 1: System Blueprint & Project Planning

COS40007 Artificial Intelligence Engineering

Dimitrios Stavropoulos - 102101565
Kai Broomfield - 103995086
Heath Sullivan - 105371303

Cl03 - G02

GitHub. Repository:
Link: https://github.com/COS40007-2026-Classrooms/CL03-G02.git

Acknowledgement of Country
We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

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1. Group Agreement & Project Plan

Project Overview

This project uses AI engineering ideas to predict how much electricity will be used in three areas of Tetouan City, Morocco. We will use the Tetouan City Power Consumption dataset (52,416 records, 10-minute intervals across 2017) to create a supervised regression model that predicts energy demand at the zone level based on environmental factors like temperature, humidity, wind speed, and solar diffuse flow measurements. To make sure that the model stays high quality and easy to maintain, the system will use MLOps and DevOps methods.

Group Roles and Responsibilities

The following table outlines the roles and responsibilities of each team member, adhering to individual strengths correlating to the project tasks. It will show the key responsibilities including a Scrum master assignment for each sprint.

Fig 1.1. Roles and responsibilities table.

Member	*Role*	*Key Responsibilities*
James (Sprint 3 Scrum master)	*Data Engineer / Design Lead*	*Data collection/preprocessing including exploratory data analysis and the ability to effectively identify and engineer useful feature columns from found dataset. The lead of architecture design flow and establishing effective communication pools, report structures/layouts and the GitHub repository branch management.*
Kai (Sprint 2 Scrum master)	*DevOps / Development Lead*	*Conceptual design, system documentation including results. Checking results from the model and dataset and deciding on retraining and monitorisation. CI/CD design.*
Heath (Sprint 1 Scrum master)	*ML Engineer / Project Lead*	*Choosing an effective deep learning model that displays a high accuracy with little to no overfitting. Model evaluation, hyperparameter tuning, PCA and a report on the performance metrics. Leading the team with deadlines, group meetings/work sessions.*

Communication Strategy

Discord and Instagram are the main ways the group talks to each other about daily updates and urgent messages. We meet in person once or twice a week, and we talk to each other outside of those meetings on a regular basis. GitHub is used to keep track of all the code and project files, and Microsoft Word (which can be shared through Word's live sharing feature) is used to edit documents together.

On-Track Behaviours

This semester long project contains a large amount of work that needs to be contributed by each group member for a successful outcome. With 3 group members from different degrees (CS, Software Engineering, and Electrical Engineering), dividing the workload into an equal contribution that adheres to each of our strengths will alleviate stress and keep us on track for deadlines and deliverables this for this subject. Each of us should adopt this attitude of thinking, much like an experienced project group that can move forward with our different strengths and weaknesses. Intuitive designs and logical architectures for the AI models we develop/train will allow for readability across the team, so behaviours such as originality, explainability and the avoidance of AI tools will positively benefit the outcomes for our specific group.

Off-Track Behaviours

Neglecting the quality of work just because there are weekly deadlines for individual and group tasks, is a dangerous approach. If group members fail to communicate and suddenly present AI generated code, models, reports, diagrams, etc, a mediocre result and even failure could be a result of it. This carries into behaviours such as laziness, lack of passion, anti-social which are detrimental to team success. This semester contains varies applied projects and portfolio tasks, if our group members lack independence and the ability to time manage , we will receive a poor result.

Conflict Resolution Approach

If conflicts happen to arise throughout the timeline of this group, mitigation strategies must be implemented and treated fair and as a serious matter. Individual concerns can be expressed through the team communication channels and can be discussed within weekly group meetings. We are an equal group that respects the individual skills and talents of one another and have committed to helping each other if any technical implementation difficulties occur.

Sprint Plan

The following diagram represents a finalised sprint plan that conveys the estimated timeline of the tasks that will be carried out by the group this semester. Task status, member tasks and goals for each component of the project is outlined with an assigned followed by task requirements.

Fig 1.2. Tetouan City Power Consumption - Sprint Plan

Sprint	Dates	Goals & Deliverables	Member Tasks for each Sprint	Status
Sprint 1	Week 6 Apr 13 – Apr 15	**Applied Project Task 1** • Project planning • System blueprint • MLOps design • EDA Due: 15 Apr	• **All members** — collaborative planning • **James** — leads system blueprint diagram, monitors model and evaluates if retraining is needed. • **Heath** — leads MLOps and DevOps. Model training • **Kai** — leads EDA and data preprocessing, engineers feature columns.	In Progress
Sprint 2	Week 7–8 Apr 20 – Apr 27	**Data Pipeline & Baseline Model** • Preprocessing pipeline • Further feature column engineering • Baseline model (regression and classification) *Lead up the Applied Project Task 2*	• **James** — preprocessing pipeline (scaling, splitting, normalization) • **Kai** — feature engineering (temporal, interaction terms, binary flags for certain columns) • **Heath** — feature selection, accuracy comparison, and final selection for task 2.	Upcoming
Sprint 3	Week 9 May 4 – May 9	**Applied Project Task 2** • Advanced modelling (RF / XGBoost) • Hyperparameter tuning • Model evaluation & comparison Due: May 4	• **Heath** — Random Forest and XGBoost implementation on top of decision tree classification and SVM • **James** — hyperparameter tuning (GridSearchCV) • **Kai** — evaluation metrics, comparison report	Upcoming
Sprint 4	Week 10–11 May 11 – May 18	**Applied Project Task 3** • MLOps integration • Monitoring & drift detection • Retraining pipeline Due: May 18	• **Heath** — monitoring dashboard, drift detection logic • **Kai** — retraining trigger logic and pipeline • **James** — model versioning, GitHub Actions setup	Upcoming
Sprint 5	Week 12–13 May 25 – Jun 1	**Demo & Final Report** • Online demo / presentation (Wk 12) • Final Applied Project Report • Peer review Due: Jun 1	• **All members** — Contribute equally to final report. • **Heath** — leads final report writeup • **James** — leads live demo preparation • **Kai** — leads presentation slides	Upcoming

2. System Blueprint Diagram

The components of this project are outlined below, with details on the approach we are taking on each individual element of our AI engineering solution/model.

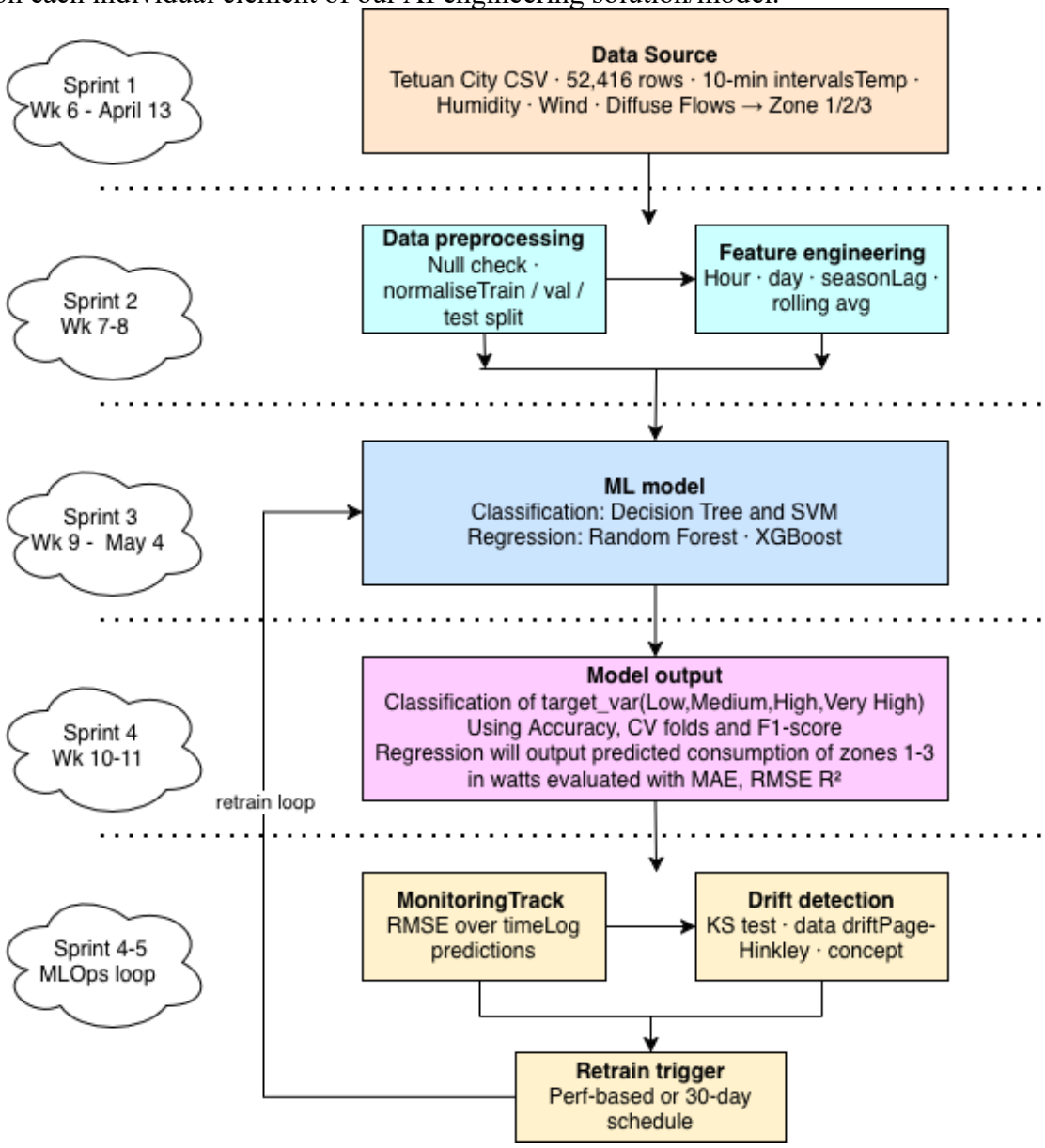


Fig 2.2. Project Component Table

Component	Description
Data Source	Tetouan City Power Consumption. Containing 52,416 rows, 10-min intervals throughout the days of Jan-Dec 2017. Features columns include DateTime, Temperature, Humidity, Wind Speed, General Diffuse Flows, Diffuse Flows. Targets: Zone 1, 2, 3 Power Consumption (measured in watts).

Data Preprocessing	DateTime split into temporal categories. StandardScalar applied to Temp, Humidity and Wind Speed. MinMaxScalar applied to the two diffuse flow categories (used for skewed data). Will also include null value checks before splitting data into a 70/20/10 train/test/val with a random seed of 42.
Feature Engineering	Temporal features extracted from the DateTime column such as Hour, Month, DayOfWeek, HourSin, HourCos. Engineered feature columns such as Temp_Humidity, Total_Solar, Temp_Hour, Peak_Hours (17-22), Night_Hours (0-6).
ML Model	Classification will use SVM model using the rbf kernal and a decision tree model. This will be trained on five engineered feature categories over five-fold cross-validation (CV) and GridSearch. Regression will make use of a tensorflow neural network to predict watt values at particular times for Zones 1-3.
Model Output	Classification model splits the target variable category (Zone 1 Power Consumption) into categories; Low, Medium, High, very High. Regression model will use predicted watt values for Zones 1-3. Performance will measure accuracy and feature ANOVA F1-Scores for classification, whilst MAE, RMSE will be used for the regression model.
Monitoring and Retraining	Monitor model accuracy, MAE, RMSE, and F1-score over evaluated cycles. Retraining will be decided if the accuracy drops below a threshold of 85%.
CI/CD	Version control via GitHub including CI/CD pipeline. We will update model versioning throughout the three sprints.

3. MLOps & DevOps Integration (High-Level Design)

3.1 Data & Training Strategy

The chosen dataset is the 2017 records of the Power Consumption of Tetouan City located in North Morocco. The dataset contains 52,417 records of 10-minute intervals and contains 6 features and 3 targets per record. The dataset features temperature, humidity, wind speed, general diffuse flows, and diffuse flows. It also contains 3 target values, Zone 1, 2, and 3

power consumption. Some additional features have also been chosen to be engineered to better suit training and testing conditions. Before training is commenced, DateTime is divided into temporal features rather than 10-minute intervals, including Hour, Month, DayOfWeek, HourSin, and HourCos to indicate cyclical patterns. Additional engineered features are Temp_Humidity, Total_solar, Temp_Hour, Peak_Hour(17:00-22:00hrs), and Night_Hours(00:00-06:00) which are derived to improve prediction patterns.
As the dataset has time-series characteristics, the training strategy will have to be a temporal split for the training/test split as the first 70% of the year and the last 30% of the year, so the training set and the test set do not have any overlap and so the model doesn't make predictions for a period it already has been trained on. For this system, two training models will be used: SVM classifier using the RBF kernel and a decision tree model. These will both be evalutated over the five-fold cross-validation with GridSearchCV for hyperparameter tuning and TensorFlow neural network for regression to predict power consumption for each zone. When retraining the models, if new datasets can be acquired to enable further training and testing, then these will be used, For a new dataset to be effective, it would need to have similar records to the current dataset and include many of the same features.

3.2 Retraining Strategy

Retraining will occur on the model if two conditions are true. The first condition is related to performance-based metrics , if the classification model accuracy drops below 85% or the regression MAE increases above a threshold of 15% on the validation set, retraining will commence. The second condition arises over time-based decisions, meaning after each sprint for this project we will conduct a results review to assess whether we need to use more data, or use a different classification/regression model more efficient for the dataset. For city data, only a 1-year (2017) dataset for an individual city can affect the model evaluation accuracy if presented with new data. To summarize, during early development retraining frequency should be low. However, if the system is constantly evolving and live data is presented to it during the later weeks of the semester , further training on new data with new features must be implemented so the logic can scale accordingly.

The retraining pipeline will go as follows:
1. Get latest validated data (different cities, in-depth, different years, etc).
2. Re-run the data preprocessing pipeline including any new feature categories that have a higher F1-Score or engineer new feature columns.
3. Train model using an extended training window (higher epochs, more folds).
4. Evaluate model on validation set with more instances.
5. Integrate new model if performance outperforms previous model.

3.3 Monitoring & Drift Detection

Monitoring the model will be achieved through the observation of the metrics; F1-score, precision, recall for the classification model, and MAE, RMSE, and R2 for the regression model. The predictions will be recorded and output into a JSON/CSV file for trend analysis alongside the generated visualisations.

The approach for drift detection involves the key feature columns (Temp, Hour, Diffuse Flows) being compared against the training baseline. This data drift refers to the shifts in the statistical output distributions of these feature inputs over time. Data drift monitorisation techniques include:

- Comparing the distribution trend of the feature columns against the training set for testing.
- Flags will be placed on any feature that records a standard deviation (std) beyond a threshold using p-value scores.
- Summarising the statistics (mean, min, max, std) per feature on a constant basis.

Concept drift is when the relationship between inputs and outputs changes. For example, if a hypothetical solar farm or industrial plant was built within the city after the dataset range of 2017, the predictions from the features will predict something completely different as the city's power consumption has changed.

3.4 DevOps Practices

For this project, version control will be maintained through utilising a shared GitHub Repository, accessed by both team members, and assessors. As each team member has individual tasks, they will work on feature-specific branches, and push/pull requests will be reviewed and tested before merging. Using a GitHub repository enables simple version tracking and allows the team to monitor and review the changes throughout all the sprints and undo changes as needed. The repository structure and branch management will be overseen by the Data Engineer / Design Lead (James)
For this project, the CI/CD pipeline will be implemented using GitHub Actions so that every time a change is pushed to the main branch, Actions runs preprocessing validation, checking missing values, ensuring no errors occur, and that the output conforms to expected shapes and thresholds without it having to be done manually. These processes are currently conceptual so the team understands how automated functions will check each push and will be implemented in a later task.
CI/CD design and oversight will be undertaken by the DevOps / Development Lead (Kai).
Documentation is maintained through collaborative Microsoft Word live documents.

4. Project Management Setup

4.1 User Stories

The following user stories define the core requirements of the system from the perspective of key stakeholders. Each story follows the standard format: As a [user], I want [goal] so that [benefit].

ID	User Story	Priority	Assigned To
U S-1	As an energy analyst, I want to be able to predict how much power each city zone will use so that I can plan how to spread out the load ahead of time.	High	James
U S-2	As the manager of city operations, I want to group consumption periods into Low, Medium, High, or Very High so that I can find the times	High	Heath

	when demand is highest and plan resources accordingly.		
U S-3	As a data engineer, I want the model to automatically retrain itself when performance falls below a certain level so that predictions stay correct over time.	High	Heath
U S-4	As a stakeholder, I want clear graphs of usage patterns and model performance metrics so that I can make smart choices about infrastructure.	Medium	Kai
U S-5	As a system administrator, I want to be notified when data drift or concept drift is found so that I can investigate it before the quality of the model goes down.	Medium	Kai
U S-6	As a developer, I want GitHub to keep track of all code and model versions and have a clear branching strategy so that working together and going back are easy.	Low	All

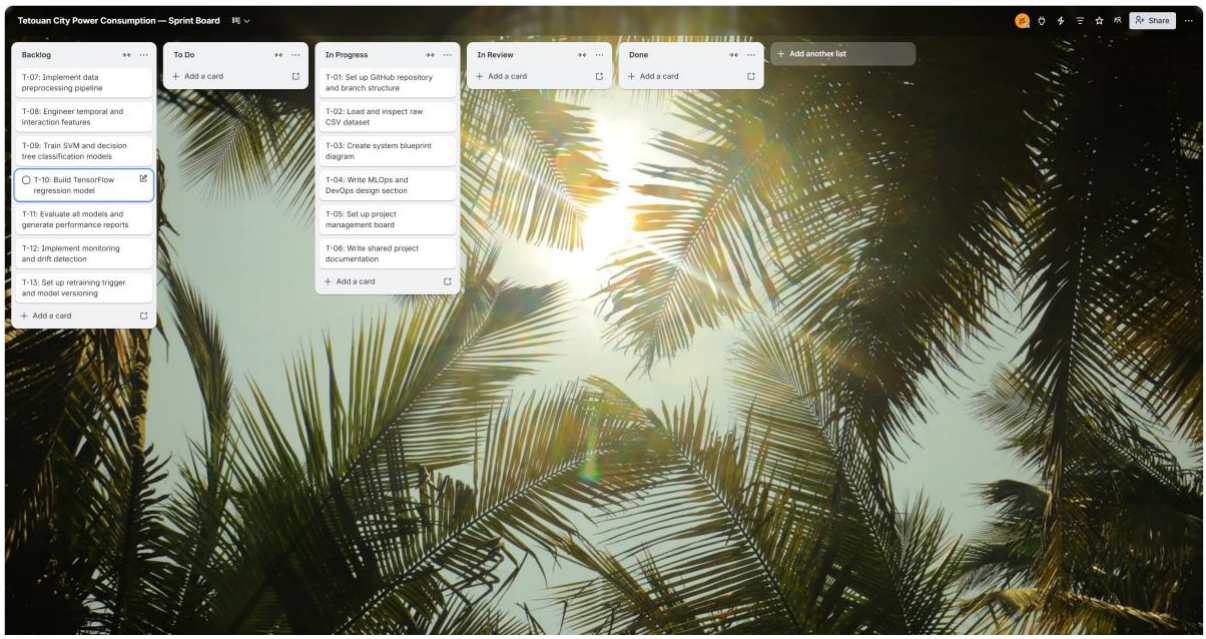
Figure 4.1. User Stories Table

4.2 Backlog & Task Breakdown

Task ID	Task Description	Assigned To	Sprint	Status
T-01	Set up GitHub repository, Branch structure, and README	Heath	Sprint 1	In progress
T-02	Load and inspect raw CSV, validate schema, check for nulls and duplicates	James	Sprint 1	In progress
T-03	Create system blueprint diagram	Heath	Sprint 1	In progress
T-04	Write MLOps and DevOps design section	Kai	Sprint 1	In progress
T-05	Set up project management board with user stories and sprint columns	Kai	Sprint 1	In progress
T-06	Write shared project documentation	James	Sprint 1	In progress
T-07	Implement data preprocessing pipeline (scaling, temporal split)	James	Sprint 2	Upcoming
T-08	Engineer temporal and interaction feature columns	James	Sprint 2	Upcoming
T-09	Train SVM and decision tree classification models with 5-fold CV and Grid Search	Heath	Sprint 2	Upcoming

4.3 Project Management Tool

The group is using a project management board with the columns "Backlog," "To Do," "In Progress," "In Review," and "Done." There are cards for each user story and its tasks, with tags for the right sprint and the right team member. Below is a screenshot of the current board and a link to the board.



Trello Link:
https://trello.com/invite/b/69ddcd9c377a5d656fc734ed/ATTIee98a5c950680861e73eec67ab9426cc677C7DEB/tetouan-city-power-consumption-sprint-board

4.4 Project Documentation

All group members can access project documentation in a shared Notion workspace. The documentation has a description of the project and its goals, an overview of the dataset and the first EDA findings, an overview of the system architecture, the main design choices and why they were made, notes from the sprint retrospective, and the roles of each team member. You can get to the Notion page by clicking on the link below.

Notion Link: https://www.notion.so/Tetouan-City-Power-Consumption-Project-Documentation-342a36ddb991808292b7f6d2063dcb59?source=copy_link

5. Contribution Statement

The table below shows what each team member did for Applied Project Task 1. All members say that the information is correct and shows how they really participated.

Member	Student Number	Contributions	%
James	102101565	Checking out the dataset, doing exploratory data analysis (EDA), designing feature engineering, making a blueprint diagram, setting up GitHub, writing project documentation, and writing a group agreement	33.3%
Kai	103995086	MLOps and DevOps design, CI/CD pipeline concept, monitoring and drift detection design, project management board setup, and user stories	33.3%
Heath	105371303	Choosing an ML model, planning a sprint, keeping track of deadlines, designing an SVM and decision tree, making a blueprint component table, and coming up with a retraining strategy	33.3%

GenAI Declaration

" Claude (Anthropic) was used to assist with explain GitHub Actions and how it can be used with this project. AI was also used to guide the Trello setup. All final code, interpretations, and conclusions are our own work after verification."

AI Evidence 1

University AI group project setup

what do i do for 4.3 regarding trello

You have two options:

Option 1 — Set up a quick Trello board (recommended) Takes about 10 minutes:

1. Go to trello.com and create a free account
2. Create a new board called **"Tetouan City Power Consumption — Sprint Board"**
3. Add these lists (columns): **Backlog → To Do → In Progress → In Review → Done**
4. Add cards for your Sprint 1 tasks (T-01 to T-06) from your task breakdown table — just the task name is enough
5. Move the ones you've done into **In Progress** or **Done**
6. Take a screenshot
7. Copy the board link (Share button → Create link)

AI Evidence 2

what is github actions

GitHub Actions is a CI/CD (Continuous Integration / Continuous Delivery) automation tool built directly into GitHub. It lets you define automated workflows that run in response to events in your repository.

How it works: You write workflow files in YAML (stored in `.github/workflows/`) that define what to do and when to do it.

Common triggers:

- A push to a branch
- A pull request being opened
- A scheduled time (cron)
- Manual trigger

What you can automate:

- Running tests whenever code is pushed
- Linting and code quality checks
- Building and packaging your application
- Deploying to a server or cloud platform
- Retraining an ML model on a schedule

Basic example:

```yaml
name: Run Tests

on: [push]

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Install dependencies
        run: pip install -r requirements.txt
      - name: Run tests
        run: pytest
```

This would automatically run your test suite every time anyone pushes code to the repo.

For your project specifically, the idea would be that whenever a change is pushed to the main branch, GitHub Actions could automatically run your preprocessing validation, check that the training script doesn't error out, and verify that model metrics meet the expected thresholds — without anyone having to manually run those checks.

It's free for public repositories and has a generous free tier for private ones.

GitHub Repository

Link: https://github.com/COS40007-2026-Classrooms/CL03-G02.git